

**BACHELOR OF COMPUTER SCIENCE
FACULTY/SCHOOL OF COMPUTER SCIENCE
BINA NUSANTARA UNIVERSITY
JAKARTA**

ASSESSMENT FORM

Course: MATH6183001 - Scientific Computing

Method of Assessment: Case Study

Semester/Academic Year : 2/2025-2026

Name of Lecturer :

Date :

Class : All

Topic : Regression & Interpolation, Taylor Series, Root of Equations, Differentiation and Integration

Group Members :	1
	2
	3
	4
	5
	6
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	8

Student Outcomes:

(SO 1) Mampu menganalisis masalah komputasi yang kompleks dan mengaplikasikan prinsip komputasi dan keilmuan lain yang sesuai untuk mengidentifikasi solusi.

Able to analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions

Learning Objectives:

(LObj 1.1) Mampu menganalisis masalah komputasi yang kompleks

Able to analyze a complex computing problem

(LObj 1.2) Mampu menerapkan prinsip komputasi dan disiplin ilmu terkait lainnya untuk mengidentifikasi solusi

Able to apply principles of computing and other relevant disciplines to identify solutions

Learning Outcomes :

(LO 1) konsep dasar dan penerapan metode numerik melalui komputasi saintifik

Explain basic concept and application of numerical method through scientific computation

(LO 2) Menyelesaikan permasalahan kuantitatif metode numerik melalui komputasi saintifik

Solve quantitative problems of numerical methods through scientific computation

(LO 3) Melakukan komputasi saintifik dasar menggunakan bahasa pemrograman Python

Compute basic scientific computation using Python programming language

(LO 4) Menganalisis hasil percobaan yang direncanakan untuk menyusun laporan ilmiah

Analyze the results of the planned experiments to create scientific reports

No	Related LO-LOBJ-SO	Assessment criteria	Weight	Excellent (85 - 100)	Good (75-84)	Average (65-74)	Poor (0 - 64)	Score	(Score x Weight)
1	LO1-L.Obj.1.1-SO1	Ability to solve regression and interpolation problems through scientific computation	20%	Able to solve in both numerical and computational approaches well without errors.	Able to solve both in numerical and computational approaches well with some errors.	Able to solve either in numerical or computational approaches well with some errors.	Unable to solve in either numerical nor computational approaches.	100	20
2	LO3-L.Obj.1.1-SO1	Ability to examine the application of the Taylors Series in scientific computation.	20%	The examination result is accurate, clear and complete.	The examination result is accurate, clear but incomplete.	The examination result is accurate, but not clear and incomplete.	The examination result is not accurate, not clear, and incomplete.	100	20

No	Related LO-LOBJ-SO	Assessment criteria	Weight	Excellent (85 - 100)	Good (75-84)	Average (65-74)	Poor (0 - 64)	Score	(Score x Weight)
3	LO2-L.Obj.1.1 -SO1 LO2-L.Obj.1.2 -SO1	Ability to examine the application of root equations in scientific computation.	20%	The examination result is accurate, clear and complete.	The examination result is accurate, clear but incomplete.	The examination result is accurate, but not clear and incomplete.	The examination result is not accurate, not clear, and incomplete.	100	20
4	LO2-L.Obj.1.1 -SO1 LO2-L.Obj.1.2 -SO1	Ability to examine the application of Numerical Differentiation and Numerical Integration in scientific computation.	20%	The examination result is accurate, clear and complete.	The examination result is accurate, clear but incomplete.	The examination result is accurate, but not clear and incomplete.	The examination result is not accurate, not clear, and incomplete.	100	20
5	LO4-L.Obj.1.1 -SO1	Ability to compute basic Python programming in assigning appropriate variables, data structure, function, statements and libraries to create scientific report	20%	The computation is correct without errors, and the report is completely written in IMRaD structure	The computation can work well with some errors, but the the report is written with minor incompleteness	The computation is correct but cannot work well, and the the report is written with minor incompleteness	The computation is incorrect and cannot work well, and report is written with major incompleteness	100	20
Total Score: $\sum(\text{Score} \times \text{Weight})$									100

Remarks:

ASSESSMENT METHOD

Instructions for Students:

1. Please read **question number 5** first before working on the other problems!
2. Submit the answer in **one .PDF document**. If you need to attach the Python code, it is recommended to use Google Colab and attach the URL to your Colab Notebook
3. All numerical answer in floating point must be provided as three-digits precision ($\times 10^{-3}$).
4. The usage of any Python libraries that provide direct function to solve a problem is allowed, however, answer with minimum uses of such libraries are rated higher.
5. This case study is consist of several open questions, which mean that each question can be answered by using different method. You'll need to decide yourself which method is suitable for each question. You **MUST** provide the reason on choosing the method in your answer.
6. You are encouraged to provide a graph to explain your answer.
7. **DO NOT** use hand-writing in any part of your answer.
8. Your answer will be checked for similarity with other student's answer. Any similarity greater than 20% (for each answer) will result in automatic 0 (zero) points.

Case Study:

You are working as a Data Scientist in EGIER, an outdoor equipment manufacturer in Bandung. As a Data Scientist, you have been tasked with processing some data on EGIER warehouse capacity. You are given a set of data that consist of the **biweekly** production of a certain type of bag they produced. The data span from January 2018 to December 2023 (given as M1 to M144). The data is given in CSV (comma-separated value) document, and you can access the document here: <https://docs.google.com/spreadsheets/d/1iSg4bEQkfi2o58HmcMDII0i6InqnZrT/edit?usp=sharing&ouid=103170622443612923929&rtpof=true&sd=true> . Your supervisor has given you a series of task that needs to be done on the data, as a part or your job!

1. **(LO1 20 points)** You must find the trend on the bag's production from the data. You must provide a mathematical model that can explain the production's trend accurately. Since your supervisor want an accurate model, you must avoid any linear approach to build the trend model.
2. **(LO3 20 points)** Since you'll need to process the data using a computer program, you'll need to convert the mathematical model from problem #1 to its numerical form (approximation). This is done so that the mathematical model can be calculated by the program easily. Since accuracy is still important, make sure that your conversion is accurate as possible. Provide an explanation to your supervisor about the accuracy of your conversion.

3. **(LO2 20 points)** The warehouse was designed to be able to store a maximum of 25,000 (twenty five thousands) bags at each month. Your supervisor asked you to provide a prediction when do EGIER need to build a new warehouse based on the trend that you have acquired in problem #2. To build a new warehouse, it is predicted that they need at least 13 months. So provide the time when EGIER need to start building their new warehouse. (**Hint:** this can be approached as a root of equation problem).
4. **(LO2 – 20 points)** Your supervisor also wants to analyze the rate of change and the cumulative production of the bags:
 - a. Estimate the monthly rate of change of bag production for the years 2020 and 2021. Identify the month with the steepest increase and steepest decrease in production, using Numerical Differentiation.
 - b. Set aside the fact that you can calculate the total production by the RAW data, Calculate the total production from January 2018 to December 2023, and compare it with the actual sum of the dataset. Explain the accuracy of your chosen Numerical Integration method (e.g., Trapezoidal Rule, Simpson's Rule, or other methods)
5. **(LO4 – 20 points)** Prepare all your answers for Problems #1–#4 in the form of a scientific report in **maximum of 3 sheets of paper (1-sided)**. Your supervisor will evaluate the clarity, completeness, and scientific quality of your report. The report should include the following sections using IMRaD structure :
 - a. Title and short Abstract
 - b. Introduction (background and objectives)
 - c. Methods (explanation of models, numerical approaches, and code used)
 - d. Results and Discussion (analysis of the obtained results, including graphs, tables, and error analysis if necessary)
 - e. References (if any)

Note for Lecturers:

1. The lecturers are advised to assess student's understanding towards the topics included in the assignment: Regression and Interpolation, Taylor Series, Root of Equation, Numerical Differentiation, and Numerical Integration
2. Provide evaluation for each question separately.
3. Refers to the given rubric to evaluate the student's answer.
4. Each question can be solve by using different methods. Please evaluate the logic and process to reach the result, not the result itself.